Systems of Linear Equations (AA HL)

ANSWER KEY

Section 1: Augmented Matrices and Gaussian Elimination

1 In row-reducing an augmented matrix, which row operation is NOT permitted?

a. Squaring all elements in a row ✓

b. Swapping two rows

c. Multiplying a row by a non-zero scalar

d. Adding a multiple of one row to another row

The three valid elementary row operations are row swaps, multiplying a row by a non-zero constant, and adding a scalar multiple of a row to another row. Squaring entries alters the solution space.

2 An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 5]$. What does this indicate about the system?

a. The system is inconsistent (has no solutions) ✓

b. The system has infinitely many solutions

c. The system has a unique solution with $z = 5$

d. The system has trivial solutions only

The row represents the contradictory equation $0x + 0y + 0z = 5$ (i.e. $0 = 5$), which cannot be satisfied by any real values; hence the system is inconsistent.

3 An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 0]$ with 2 leading pivots across 3 variables. What does this signify?

a. The system has infinitely many solutions parameterized by 1 free variable ✓

b. The system has a unique solution

c. The system has no solutions

d. The determinant of the matrix is 1

With 3 variables and 2 leading pivot columns, there is $3 - 2 = 1$ free variable, which gives a line of infinitely many solutions.

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Section 2: Geometric Interpretation of 3 x 3 Systems

4 What is the geometric configuration of three linear equations in three variables that possess a unique solution?

a. **Three planes intersecting at a single unique point ✓**

b. Three planes intersecting along a common line

c. Three parallel non-intersecting planes

d. Three planes forming a triangular prism

A unique solution (x_0, y_0, z_0) means all three planes share exactly one point of intersection.

5 If three planes intersect along a single straight line, what type of system is this?

a. **Consistent dependent system with infinitely many solutions ✓**

b. Inconsistent system with no solutions

c. Consistent independent system with 1 solution

d. Homogeneous system with trivial solution only

Intersection along a line means infinitely many common points exist, parameterized by 1 free variable.

6 Three planes form the faces of an open triangular prism with no common line or point. How is this system classified?

a. **Inconsistent (no solutions) ✓**

b. Infinitely many solutions

c. Unique solution

d. Singular dependent

Each pair of planes intersects in a line, but all three lines are parallel and distinct, meaning there is no point common to all three planes simultaneously.

Section 3: Determinants and Cramer's Rule

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7 Evaluate the 2×2 determinant $\det[[4, -3], [2, 5]]$.

a. 26 ✓

b. 14

c. -26

d. 20

$\det = ad - bc = 4(5) - (-3)(2) = 20 - (-6) = 26$.

8 Under what condition does a 3×3 system $A\mathbf{x} = \mathbf{b}$ have a unique solution?

a. $\det(A) \neq 0$ ✓

b. $\det(A) = 0$

c. $\mathbf{b} = \mathbf{0}$

d. $\text{Tr}(A) > 0$

A square system $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if the coefficient matrix A is non-singular, meaning $\det(A) \neq 0$.

9 If $\det(A) = 0$ for a system $A\mathbf{x} = \mathbf{b}$, what can be concluded?

a. **The system either has no solutions or infinitely many solutions** ✓

b. The system must have no solutions

c. The system must have infinitely many solutions

d. The system has exactly 2 solutions

When $\det(A) = 0$, A is singular; depending on the consistency of the constant vector $\mathbf{b}$, the system has either zero solutions or infinitely many solutions.

Section 4: Parameterized Solutions and Free Variables

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10 Solve the system: $x + y + z = 6$ and $2x - y + z = 3$. Parameterize the solution line using $z = \lambda$.

a. $x = 3 - \frac{2}{3}\lambda, y = 3 - \frac{1}{3}\lambda, z = \lambda$ ✓

b. $x = 3 + \lambda, y = 3 - \lambda, z = \lambda$

c. $x = 9 - 2\lambda, y = \lambda, z = \lambda$

d. $x = 1, y = 2, z = 3$

Add equations: $3x + 2z = 9 \Rightarrow 3x = 9 - 2\lambda \Rightarrow x = 3 - \frac{2}{3}\lambda$. Subtract from first: $y = 6 - x - z = 6 - (3 - \frac{2}{3}\lambda) - \lambda = 3 - \frac{1}{3}\lambda$.

11 For what value of k does the system $x + 2y = 4$ and $3x + 6y = k$ have infinitely many solutions?

a. $k = 12$ ✓

b. $k = 4$

c. $k = 0$

d. $k = 6$

Multiply the first equation by 3: $3x + 6y = 12$. For infinitely many solutions, the equations must be identical, so $k = 12$.

12 If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions (solutions other than $\mathbf{x} = \mathbf{0}$), what must be true about $\det(A)$?

a. $\det(A) = 0$ ✓

b. $\det(A) > 0$

c. $\det(A) = 1$

d. $\det(A) < 0$

A homogeneous linear system has non-trivial solutions if and only if the coefficient matrix is singular, i.e., $\det(A) = 0$.

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